BIOR · A FOOD SAFETY GAME

You are the risk assessor

The printable game · Rules

RISK = **HOW SERIOUS** × **HOW OFTEN**

The game shows how risk assessors judge food risks, and why the real risk is often not where we expect it. Anyone who buys, cooks or eats food, from about age 10. No prior knowledge needed.

Play online too: **riskuspele.github.io**

What you need

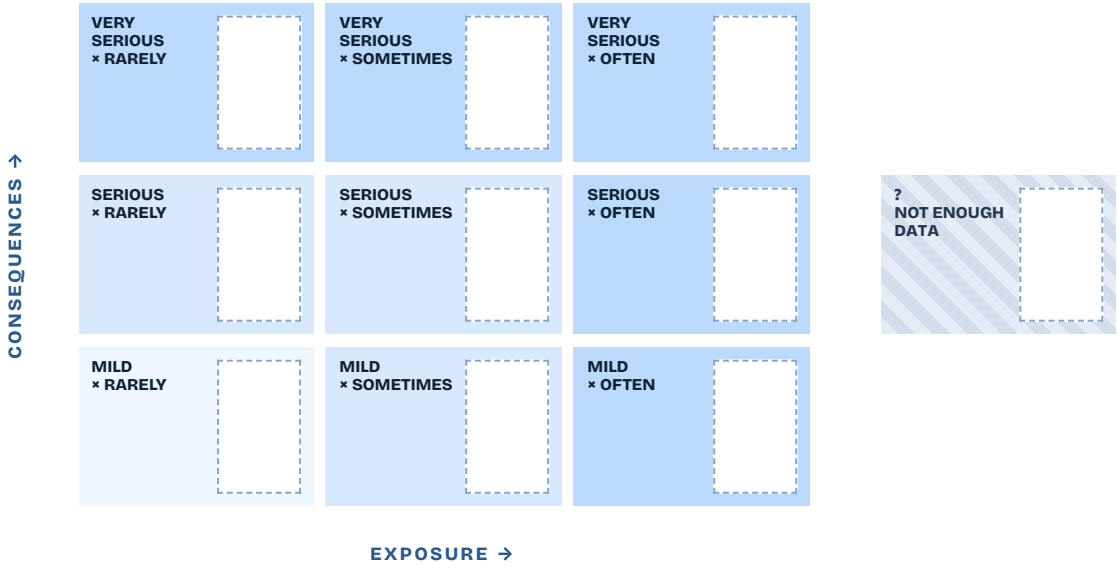
- A printer and A4 paper. Thicker paper suits the cards, 160–250 g/m².
- Scissors or a paper cutter.
- One or more players or teams.

How to print

- Print this file (rules, board and answers) single-sided, and the cheat sheets too if you want them.
- Print the cards file double-sided, flipping on the long edge. If your printer cannot print on both sides, print the odd pages, put them back in the printer the other way up and print the even pages.
- Print at 100 % (actual size), not “fit to page”.
- Cut along the dashed lines through the middle: a card sheet gives 4 cards, a board sheet 2 squares. If you like, trim the white frame off the cards along the grey line. Each card's number is on its back too, so you can check that the two sides match.

Set up the board

Lay the nine squares out on a table as in the picture: up means more serious consequences, right means more often. Put the “Not enough data” square beside them. Shuffle the cards and stack them face up.



How to play

1 Read the card.

WHAT IT DOES tells you about the consequences. WHERE IT IS FOUND, HOW OFTEN IT IS FOUND and HOW OFTEN IT IS EATEN together tell you how often and how much we meet it.

2 Put it on the square where you think it belongs.

Judge how it ends for the people it affects, at the amounts we actually take in. A rare exception does not count, but a severe outcome for one group, such as small children or pregnant women, does. On a hazard card, frequency is both lines together: how often it is found in food and how often it is eaten. If there is not enough data, put the card on “Not enough data”. One square can hold several cards.

3 Turn the cards over.

When every card is down, turn them over. The back shows where the risk assessors put them, and why.

What the rows and columns mean

CONSEQUENCES ↑ VERY SERIOUS

Can cause permanent damage or be fatal.

CONSEQUENCES ↑ SERIOUS

May need treatment or a hospital, or can harm health over the longer term.

CONSEQUENCES ↑ MILD

Usually passes on its own within days, with no lasting harm.

EXPOSURE → RARELY

Few people meet it, rarely or in small amounts.

EXPOSURE → SOMETIMES

Some people meet it now and then.

EXPOSURE → OFTEN

Almost everyone meets it regularly, even daily.

? NOT ENOUGH DATA

Put it here if there is not enough data to assess the risk

Points, if you keep score

Spot on	**100**
Next square	**50**
Diagonal square	**25**

A card that belongs on “Not enough data” scores only there. A card on “Not enough data” that belongs on the matrix scores nothing.

Each cheat sheet used takes off a quarter of the card's points.

Tips

- Start with five or six cards, not all of them at once.
- Teams first agree on a square and say why, and only then turn the card over.
- If a card seems too hard, look its number up in the cheat sheets.
- Set 1 (1–44) holds the cards from BIOR's stand, set 2 (45–76) the new ones.

Each card's square is based on published risk assessments: the Finnish Food Authority (Ruokavirasto 1/2021), the Norwegian Scientific Committee for Food and Environment (VKM 2019:13, 2021:10), EFSA and ECDC (2024), and the Technical University of Denmark (DTU, 2023). The frame of reference is a person in Europe, Latvia included, today. They are well-founded starting points for a conversation, not a formal risk assessment.

Game by Jānis Ruško, BIOR. The game was made within research application No. 1.1.1.9/LZP/1/24/124 “From Data to Decisions: Smart Tools for Food Safety Risk Assessment and Prioritization” at the Institute of Food Safety, Animal Health and Environment “BIOR”.

Illustrations on some of the cards: Elīna Blekte-Gerāne, from the EFSA #Safe2EatEU campaign activity book “Kur slēpjas Baktiņi?” (2024), run in Latvia by BIOR. Used with permission. All other drawings and the chemical structures were made for this game.

Co-funded by the European Union

CONSEQUENCES ↑

VERY SERIOUS

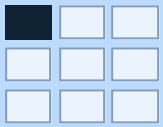
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✖

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CARDS HERE

CONSEQUENCES ↑

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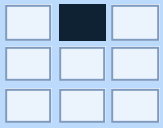
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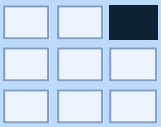
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EXPOSURE →

OFTEN

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CARDS HERE

CONSEQUENCES ↑

SERIOUS

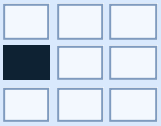
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CONSEQUENCES ↑

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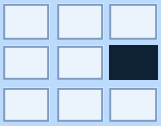
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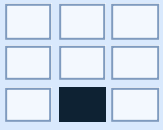
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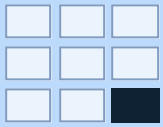
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CARDS HERE

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NOT ENOUGH DATA

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How well do we know it?

CARDS HERE

Answers

Where the risk assessors put each card. The back of the card says why.

No.	Card	Square
01	Campylobacter	Serious × Sometimes
02	Salmonella	Serious × Rarely
03	Listeria monocytogenes	Very serious × Rarely
04	Norovirus	Mild × Often
05	Toxoplasma gondii	Very serious × Sometimes
06	Shiga toxin-producing E. coli	Very serious × Rarely
07	Yersinia enterocolitica	Serious × Rarely
08	Aflatoxins	Very serious × Rarely
09	Deoxynivalenol (DON)	Mild × Often
10	T-2 and HT-2 toxins	Serious × Sometimes
11	Ochratoxin A	Serious × Sometimes
12	Patulin	Mild × Rarely
13	Alternaria toxins	Serious × Often
14	Acrylamide	Serious × Often
15	Furan	Serious × Sometimes
16	Benzo(a)pyrene and PAHs	Serious × Sometimes
17	Glycidyl esters	Serious × Sometimes
18	Lead	Serious × Often
19	Cadmium	Mild × Often
20	Inorganic arsenic	Very serious × Sometimes
21	Methylmercury	Serious × Rarely
22	PFAS (“forever chemicals”)	Serious × Often
23	Dioxins and dl-PCBs	Serious × Often
24	Histamine	Mild × Rarely
25	Nitrate	Mild × Often
26	Pesticide residues	Mild × Sometimes
27	Glutamate (E621)	Mild × Often
28	Peanut allergy	Very serious × Rarely
29	Aspartame	Mild × Sometimes
30	Microplastics	Not enough data
31	Titanium dioxide (E171)	Not enough data
32	Cyanogenic glycosides	Mild × Rarely
33	Undercooked chicken	Serious × Often
34	Home-slaughtered game	Very serious × Rarely
35	One board for meat and salad	Serious × Often
36	Unwashed hands	Mild × Often
37	Cooked too brown	Serious × Often
38	Washing chicken before cooking	Serious × Sometimes
39	Unwashed salad and berries	Mild × Often
40	Frozen berries straight from the freezer	Mild × Sometimes
41	Raw milk	Very serious × Rarely
42	Mushrooms you picked yourself	Very serious × Rarely
43	Undercooked burger	Very serious × Sometimes
44	Energy drinks	Not enough data
45	Leftovers left out	Mild × Sometimes
46	Thawing meat on the counter	Mild × Often
47	Frozen breaded chicken	Serious × Rarely
48	Frozen vegetables, uncooked	Very serious × Rarely
49	Raw dough	Very serious × Sometimes
50	Eating food past its “use by” date	Very serious × Rarely
51	Eating food past its “best before” date	Mild × Often
52	Scraping off the mould	Not enough data
53	Charred barbecue meat	Serious × Sometimes
54	Several supplements at once	Very serious × Rarely
55	Nitrosamines	Serious × Often
56	Ergot alkaloids	Serious × Sometimes
57	Pyrrolizidine alkaloids	Very serious × Sometimes
58	Tropane alkaloids	Serious × Rarely
59	Lead from petrol	Very serious × Often
60	Methylmercury in Minamata	Very serious × Often
61	Cadmium in Toyama rice	Very serious × Often
62	Melamine in infant formula	Very serious × Often
63	Ergotism epidemic	Very serious × Often
64	Polyols (sugar alcohols)	Mild × Sometimes
65	Honey for babies	Very serious × Rarely
66	Home-canned food	Very serious × Rarely
67	Hepatitis E virus	Mild × Sometimes
68	Anisakis: fish parasites	Serious × Rarely
69	Chemicals from packaging and tableware	Mild × Often
70	6PPD-quinone	Not enough data
71	Cinnamon and coumarin	Mild × Sometimes
72	Well water	Serious × Sometimes
73	Microwaves	Mild × Often
74	Swallowed chewing gum	Mild × Sometimes
75	Swallowed watermelon seeds	Mild × Sometimes
76	Swimming straight after eating	Mild × Sometimes